

Day 3: EC2 Deep Dive & Infrastructure Storage

Instance Families, Keypairs, EBS, Instance Store, and Security Groups

Module 1: EC2 Families & Access Control

AWS categorizes compute virtual instances into multiple instance families to address different performance profiles. Administrators must select the appropriate family to optimize runtime metrics and reduce operating costs.

Family	Example	Resource Characteristics	Target Use Cases
General Purpose	t3.micro, m5	Balanced CPU, memory, and networks	Web servers, code repositories, testing platforms
Compute Optimized	c5, c6g	High vCPU-to-memory allocation ratios	High-traffic web backends, batch scripts, media encoders
Memory Optimized	r5, r6g	Large RAM allocation per vCPU	In-memory database engines (Redis), relational DBs, analytics
Storage Optimized	i3, d2	High disk I/O rates, large local storage	NoSQL systems, data warehouses, distributed filesystems
Accelerated Compute	p3, g4	Equipped with dedicated GPUs/ASICs	Machine learning models, graphics rendering, calculations

Key Pairs & Security Groups

- **Key Pairs (SSH Access):** Asymmetric cryptography secures access to EC2. The public key is loaded into the instance during launch, while you save the private key file (.pem). Connection requires authentication matching.
- **Security Groups:** Stateful virtual firewalls that control traffic at the instance interface level. Stateful behavior automatically allows outbound responses to approved inbound connections.

Module 2: Cloud Storage & Command Line Copying

Elastic Block Store (EBS) vs. Instance Store

AWS instances utilize block storage solutions with distinct lifecycles. EBS represents virtual disks, while Instance Store represents hardware physically attached to the motherboard hosting the VM.

Attribute	Elastic Block Store (EBS)	Instance Store
Storage Type	Network-attached Virtual Volume	Direct-attached physical SSD/HDD
Persistence	Persistent. Data survives instance stop/start	Ephemeral. Data is destroyed on stop/termination
Performance	Subject to network speeds; customizable size	Ultra-fast read/write throughput; fixed size

Attribute	Elastic Block Store (EBS)	Instance Store
Backups	Easy snapshots saved directly to Amazon S3	No automated backups; must duplicate data manually

File Transfers via Git Bash (SCP command)

To securely transfer files from a local windows machine to an EC2 instance, use the Secure Copy Protocol:

```
scp -i /path/to/key.pem /path/to/local/file.html  
ubuntu@ec2-ip-address:/remote/path/
```